

Learning Criterion

C4: Applies the Minimax Regret criterion to make decisions without probabilities, selecting the alternative with the smallest possible maximum regret.

1 Minimax Regret

When probabilities are unavailable, the Minimax Regret criterion compares the regret associated with each alternative under each possible state of nature.

Focus	Maximum possible regret
First step	Find the best payoff in each state of nature
Second step	Construct the regret table
Third step	Find the maximum regret for each alternative
Final step	Choose the smallest maximum regret

First, identify the best payoff in each state of nature:

$$b(S_j) = \max_i P_{ay}(A_i, S_j)$$

Then calculate the regret for each alternative and state:

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$$

For each alternative, find its maximum regret:

$$R(A_i) = \max_j r(A_i, S_j)$$

Finally, choose the alternative with the smallest maximum regret:

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i)$$

2 Solved Example: Clothing Store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager knows that customer response to the campaign is uncertain.

S_1 : Strong Sales S_2 : Weak Sales

Alternative	S_1	S_2
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars. What decision should the manager make using the Minimax Regret criterion?

2.1 Solution

Step 1: Identify the best payoff in each state of nature

$$\text{Under strong sales: } b(S_1) = \max\{55, 35\} = 55$$

$$\text{Under weak sales: } b(S_2) = \max\{5, 22\} = 22$$

Step 2: Construct the opportunity loss table

Regret is calculated as

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1 :	$r(A_1, S_1) = 55 - 55 = 0$	$r(A_1, S_2) = 22 - 5 = 17$
For A_2 :	$r(A_2, S_1) = 55 - 35 = 20$	$r(A_2, S_2) = 22 - 22 = 0$

Step 3: Find the maximum regret for each alternative

$$R(A_1) = \max\{0, 17\} = 17$$

$$R(A_2) = \max\{20, 0\} = 20$$

Step 4: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{17, 20\} = 17$$

Thus,

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_1$$

The Minimax Regret criterion recommends launching the new campaign because its maximum possible regret is \$17,000, compared with \$20,000 for keeping the current strategy.

Extended opportunity loss table

Alternative	S_1	S_2	Maximum Regret
A_1	0	17	$\max\{0, 17\} = 17$
A_2	20	0	$\max\{20, 0\} = 20$
Minimum Maximum Regret			$\min\{17, 20\} = 17$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_1$$

3 Guided Practice: Sports Equipment Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low. The company has no reliable information about the probability of each demand level.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	90	30	-15
Medium production	65	45	20
Small production	45	38	28
Maintain production	20	20	20

All payoffs are expressed in thousands of dollars. Determine the decision using the Minimax Regret criterion.

Step 1: Identify the best payoff in each state of nature

$$b(S_1) = \max\{90, 65, 45, 20\} = \underline{\hspace{2cm}}$$

$$b(S_2) = \max\{30, 45, 38, 20\} = \underline{\hspace{2cm}}$$

$$b(S_3) = \max\{-15, 20, 28, 20\} = \underline{\hspace{2cm}}$$

Step 2: Construct the opportunity loss table

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1 : $r(A_1, S_1) = \underline{\hspace{2cm}}$ $r(A_1, S_2) = \underline{\hspace{2cm}}$ $r(A_1, S_3) = \underline{\hspace{2cm}}$
For A_2 : $r(A_2, S_1) = \underline{\hspace{2cm}}$ $r(A_2, S_2) = \underline{\hspace{2cm}}$ $r(A_2, S_3) = \underline{\hspace{2cm}}$
For A_3 : $r(A_3, S_1) = \underline{\hspace{2cm}}$ $r(A_3, S_2) = \underline{\hspace{2cm}}$ $r(A_3, S_3) = \underline{\hspace{2cm}}$
For A_4 : $r(A_4, S_1) = \underline{\hspace{2cm}}$ $r(A_4, S_2) = \underline{\hspace{2cm}}$ $r(A_4, S_3) = \underline{\hspace{2cm}}$

Therefore, complete the opportunity loss table:

Alternative	S_1	S_2	S_3
A_1	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
A_2	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
A_3	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
A_4	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$

Step 3: Find the maximum regret for each alternative

$$R(A_1) = \max\{\underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}\} = \underline{\hspace{2cm}}$$

$$R(A_2) = \max\{\underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}\} = \underline{\hspace{2cm}}$$

$$R(A_3) = \max\{\underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}\} = \underline{\hspace{2cm}}$$

$$R(A_4) = \max\{\underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}\} = \underline{\hspace{2cm}}$$

Step 4: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{\underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}\} = \underline{\hspace{2cm}}$$

Thus,

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = \underline{\hspace{2cm}}$$

4 Independent Practice: Furniture Company

A furniture company must choose a production strategy for a new collection. Demand could be high, medium, or low.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	95	28	-18
Medium production	70	50	16
Small production	48	40	32
Maintain production	25	25	25

Step 1: Find the best payoff in each state

$$b(S_1) = \max\{95, 70, 48, 25\} = \underline{\hspace{2cm}}$$

$$b(S_2) = \max\{28, 50, 40, 25\} = \underline{\hspace{2cm}}$$

$$b(S_3) = \max\{-18, 16, 32, 25\} = \underline{\hspace{2cm}}$$

Step 2: Complete the opportunity loss table

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_3	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_4	<u> </u>	<u> </u>	<u> </u>	<u> </u>

Step 3: Apply the Minimax Regret criterion

$$R^* = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}}$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = \underline{\hspace{2cm}}$$

5 Independent Practice: Electronics Company

An electronics company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	88	32	-12
Medium production	64	46	14
Small production	42	38	30
Maintain production	20	20	20

Step 1: Find the best payoff in each state

$$b(S_1) = \max\{88, 64, 42, 20\} = \underline{\hspace{2cm}}$$

$$b(S_2) = \max\{32, 46, 38, 20\} = \underline{\hspace{2cm}}$$

$$b(S_3) = \max\{-12, 14, 30, 20\} = \underline{\hspace{2cm}}$$

Step 2: Complete the opportunity loss table

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_3	<u> </u>	<u> </u>	<u> </u>	<u> </u>
A_4	<u> </u>	<u> </u>	<u> </u>	<u> </u>

Step 3: Apply the Minimax Regret criterion

$$R^* = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}}$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = \underline{\hspace{2cm}}$$

6 Independent Practice: Coffee Shop

A coffee shop must decide how many pastries to prepare each morning. Demand could be high, medium, or low.

- A_1 : Prepare a large quantity of pastries.
- A_2 : Prepare a medium quantity of pastries.
- A_3 : Prepare a small quantity of pastries.
- A_4 : Maintain the usual quantity of pastries.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	72	38	8
A_2	58	45	20
A_3	40	35	28
A_4	22	22	22

What alternative should the coffee shop choose using the Minimax Regret criterion?

7 Independent Practice: Delivery Company

A delivery company must choose a fleet size for the holiday season. Customer demand could be high, medium, or low.

- A_1 : Operate a large fleet.
- A_2 : Operate a medium fleet.
- A_3 : Operate a small fleet.
- A_4 : Maintain the current fleet size.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	105	42	-20
A_2	82	55	12
A_3	60	48	30
A_4	35	35	35

What alternative should the delivery company choose using the Minimax Regret criterion?

8 Independent Practice: Clothing Store

A clothing store must decide how many winter jackets to order before the season begins. Demand could be high, medium, or low.

- A_1 : Place a large order of winter jackets.
- A_2 : Place a medium order of winter jackets.
- A_3 : Place a small order of winter jackets.
- A_4 : Maintain the current inventory level.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	90	34	-15
A_2	68	52	18
A_3	46	42	32
A_4	25	25	25

What alternative should the clothing store choose using the Minimax Regret criterion?

9 Independent Practice: Food Truck

A food truck must decide how much food to prepare for a weekend event. The number of customers could be high, medium, or low.

- A_1 : Prepare a large amount of food.
- A_2 : Prepare a medium amount of food.
- A_3 : Prepare a small amount of food.
- A_4 : Maintain the usual amount of food.

Alternative	Number of Customers		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	84	30	-10
A_2	63	49	16
A_3	44	41	29
A_4	24	24	24

What alternative should the food truck choose using the Minimax Regret criterion?

SOLUTIONS

10 Guided Practice: Sports Equipment Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low. The company has no reliable information about the probability of each demand level.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	90	30	-15
Medium production	65	45	20
Small production	45	38	28
Maintain production	20	20	20

All payoffs are expressed in thousands of dollars. Determine the decision using the Minimax Regret criterion.

Step 1: Identify the best payoff in each state of nature

$$\begin{aligned}
 b(S_1) &= \max\{90, 65, 45, 20\} = 90 \\
 b(S_2) &= \max\{30, 45, 38, 20\} = 45 \\
 b(S_3) &= \max\{-15, 20, 28, 20\} = 28
 \end{aligned}$$

Step 2: Construct the opportunity loss table

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For A_1 :	$r(A_1, S_1) = 90 - 90 = 0$
	$r(A_1, S_2) = 45 - 30 = 15$
	$r(A_1, S_3) = 28 - (-15) = 43$
For A_2 :	$r(A_2, S_1) = 90 - 65 = 25$
	$r(A_2, S_2) = 45 - 45 = 0$
	$r(A_2, S_3) = 28 - 20 = 8$
For A_3 :	$r(A_3, S_1) = 90 - 45 = 45$
	$r(A_3, S_2) = 45 - 38 = 7$
	$r(A_3, S_3) = 28 - 28 = 0$
For A_4 :	$r(A_4, S_1) = 90 - 20 = 70$
	$r(A_4, S_2) = 45 - 20 = 25$
	$r(A_4, S_3) = 28 - 20 = 8$

Therefore, the opportunity loss table is:

Alternative	S_1	S_2	S_3
A_1	0	15	43
A_2	25	0	8
A_3	45	7	0
A_4	70	25	8

Step 3: Find the maximum regret for each alternative

$$\begin{aligned}
 R(A_1) &= \max\{0, 15, 43\} = 43 \\
 R(A_2) &= \max\{25, 0, 8\} = 25 \\
 R(A_3) &= \max\{45, 7, 0\} = 45 \\
 R(A_4) &= \max\{70, 25, 8\} = 70
 \end{aligned}$$

Step 4: Apply the Minimax Regret criterion

Choose the alternative with the smallest maximum regret:

$$R^* = \min\{43, 25, 45, 70\} = 25$$

Thus,

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

11 Independent Practice: Furniture Company

A furniture company must choose a production strategy for a new collection. Demand could be high, medium, or low.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	95	28	-18
Medium production	70	50	16
Small production	48	40	32
Maintain production	25	25	25

Step 1: Find the best payoff in each state

$$b(S_1) = \max\{95, 70, 48, 25\} = 95$$

$$b(S_2) = \max\{28, 50, 40, 25\} = 50$$

$$b(S_3) = \max\{-18, 16, 32, 25\} = 32$$

Step 2: Complete the opportunity loss table

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	22	50	50
A_2	25	0	16	25
A_3	47	10	0	47
A_4	70	25	7	70

Step 3: Apply the Minimax Regret criterion

$$R^* = \min\{50, 25, 47, 70\} = 25$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

12 Independent Practice: Electronics Company

An electronics company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	88	32	-12
Medium production	64	46	14
Small production	42	38	30
Maintain production	20	20	20

Step 1: Find the best payoff in each state

$$b(S_1) = \max\{88, 64, 42, 20\} = 88$$

$$b(S_2) = \max\{32, 46, 38, 20\} = 46$$

$$b(S_3) = \max\{-12, 14, 30, 20\} = 30$$

Step 2: Complete the opportunity loss table

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	14	42	42
A_2	24	0	16	24
A_3	46	8	0	46
A_4	68	26	10	68

Step 3: Apply the Minimax Regret criterion

$$R^* = \min\{42, 24, 46, 68\} = 24$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

13 Independent Practice: Coffee Shop 14 Independent Practice: Delivery Company

A coffee shop must decide how many pastries to prepare each morning. Demand could be high, medium, or low.

- A_1 : Prepare a large quantity of pastries.
- A_2 : Prepare a medium quantity of pastries.
- A_3 : Prepare a small quantity of pastries.
- A_4 : Maintain the usual quantity of pastries.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	72	38	8
A_2	58	45	20
A_3	40	35	28
A_4	22	22	22

Solution

$$b(S_1) = \max\{72, 58, 40, 22\} = 72$$

$$b(S_2) = \max\{38, 45, 35, 22\} = 45$$

$$b(S_3) = \max\{8, 20, 28, 22\} = 28$$

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	7	20	20
A_2	14	0	8	14
A_3	32	10	0	32
A_4	50	23	6	50

$$R^* = \min\{20, 14, 32, 50\} = 14$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

The coffee shop should prepare a medium quantity of pastries.

A delivery company must choose a fleet size for the holiday season. Customer demand could be high, medium, or low.

- A_1 : Operate a large fleet.
- A_2 : Operate a medium fleet.
- A_3 : Operate a small fleet.
- A_4 : Maintain the current fleet size.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	105	42	-20
A_2	82	55	12
A_3	60	48	30
A_4	35	35	35

Solution

$$b(S_1) = \max\{105, 82, 60, 35\} = 105$$

$$b(S_2) = \max\{42, 55, 48, 35\} = 55$$

$$b(S_3) = \max\{-20, 12, 30, 35\} = 35$$

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	13	55	55
A_2	23	0	23	23
A_3	45	7	5	45
A_4	70	20	0	70

$$R^* = \min\{55, 23, 45, 70\} = 23$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

The delivery company should operate a medium fleet.

15 Independent Practice: Clothing Store

A clothing store must decide how many winter jackets to order before the season begins. Demand could be high, medium, or low.

- A_1 : Place a large order of winter jackets.
- A_2 : Place a medium order of winter jackets.
- A_3 : Place a small order of winter jackets.
- A_4 : Maintain the current inventory level.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	90	34	-15
A_2	68	52	18
A_3	46	42	32
A_4	25	25	25

Solution

$$\begin{aligned}
 b(S_1) &= \max\{90, 68, 46, 25\} = 90 \\
 b(S_2) &= \max\{34, 52, 42, 25\} = 52 \\
 b(S_3) &= \max\{-15, 18, 32, 25\} = 32
 \end{aligned}$$

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	18	47	47
A_2	22	0	14	22
A_3	44	10	0	44
A_4	65	27	7	65

$$R^* = \min\{47, 22, 44, 65\} = 22$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

The clothing store should place a medium order of winter jackets.

16 Independent Practice: Food Truck

A food truck must decide how much food to prepare for a weekend event. The number of customers could be high, medium, or low.

- A_1 : Prepare a large amount of food.
- A_2 : Prepare a medium amount of food.
- A_3 : Prepare a small amount of food.
- A_4 : Maintain the usual amount of food.

Alternative	Number of Customers		
	High (S_1)	Medium (S_2)	Low (S_3)
A_1	84	30	-10
A_2	63	49	16
A_3	44	41	29
A_4	24	24	24

Solution

$$\begin{aligned}
 b(S_1) &= \max\{84, 63, 44, 24\} = 84 \\
 b(S_2) &= \max\{30, 49, 41, 24\} = 49 \\
 b(S_3) &= \max\{-10, 16, 29, 24\} = 29
 \end{aligned}$$

Alternative	S_1	S_2	S_3	Maximum Regret
A_1	0	19	39	39
A_2	21	0	13	21
A_3	40	8	0	40
A_4	60	25	5	60

$$R^* = \min\{39, 21, 40, 60\} = 21$$

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i) = A_2$$

The food truck should prepare a medium amount of food.